Assignment 1: Branch Predictor Implementation

Course 6385

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1 How to run

The handout builds under `cbp2-infrastructure-v2`. In this repository that folder is the root. `src/`, `traces/`, and `run` are here. Each block below starts from that root.

Build and score one trace. The program prints one line, the mispredictions per thousand instructions.

```
make -C src
./src/predict traces/164.gzip/gzip.trace.bz2
```

Score every trace. `run` calls `./src/predict`. The trace reader decompresses each file with `/usr/bin/bzip2 -dc`.

```
make -C src
csh ./run traces
```

Check the Appendix B worksheet. The checker looks for `../examples/appendix-b` from the current directory, so it is run from `src`. A match ends with `PASS: 22 / 22 rows match the csv tables`.

```
make -C src appendix_b_check
cd src && ./appendix_b_check
```

2 What was implemented

The submitted predictor guesses whether a conditional branch is taken. `my_predictor.h` has two copies of the same idea. `toy_pm_predictor` is the Appendix B worksheet: 16 local counters, 4 global sets, 2 ways, and 4 bits of history. The checker uses only that copy. `pm_predictor` is the copy `predict.cc` runs on the traces. It is the Pentium M size: a local table, a history of recent outcomes, and a tagged global table with 4 ways.

For each conditional branch the large predictor does three things.

1. **Local guess.** A table of 4096 two-bit counters is indexed by `PC[11:0]`. Each counter counts how often that address was taken. Values 0 and 1 predict not taken. Values 2 and 3 predict taken. The counter stops at 0 and at 3.

2. **Recent outcomes.** A 15-bit register shifts in a 1 when the branch was taken and a 0 when it was not. This replaces the Pentium M path register, which the assignment does not require.

3. **Global guess.** `PC[16:2]` is XORed with those 15 history bits, the same split as Appendix B with a wider address. Bits `[14:6]` pick one of 512 sets. Bits `[5:0]` are a tag. Each set holds four entries. If one entry is valid and the tag matches, its counter makes the prediction. Otherwise the local guess is used.

2.1 Changes from the worksheet

`toy_pm_predictor` keeps the Appendix B sizes, the two-way marker that records which way was used last, and an update of both tables on every conditional branch. `pm_predictor` keeps the same prediction rule: a global hit wins, and a miss uses the local counter. The index is the same XOR split, widened from 4 bits to 15. Three things change.

- **Tables.**
  - 16 local counters become 4,096, indexed by `PC[11:0]`.
  - 4 global sets become 512, and 2 ways become 4. The tag grows from 2 bits to 6.
  - 4 bits of history become 15. `PC[16:2]` is XORed with that history. Bits `[14:6]` select the set and bits `[5:0]` are the tag.
- **Replacement.** Each of the four ways stores a rank. Rank 0 is the newest. A miss fills an empty way, or the way with the highest rank.
- **Training.**
  - The local counter is trained when the global table misses.
  - A global entry is allocated only when that local guess was wrong. A replaced counter starts at 2 if the branch was taken and at 1 if it was not.
  - A global hit that is wrong, while the local guess was right, clears that entry.

Trace counters start at zero. The worksheet copy keeps the Appendix B starting values, so its counters still match the example.

2.2 What was skipped

The loop predictor, the branch target buffer, and the path register are omitted. Unconditional branches are predicted taken and do not change the tables. Branch targets are not predicted. The score is direction misses only.

3 Check against Appendix B

Appendix B walks through 22 branches at `0x44`, `0x6C`, and `0x90`. The expected history, index, local counters, and global entries are in `hybrid.csv`, `local.csv`, and `global.csv`. `appendix_b_check` feeds each address and outcome to `toy_pm_predictor` and compares every saved field.

That run matches 22 of 22 rows. The large predictor is a different width, so it is not checked against those rows.

4 Trace results

Each of the 20 traces is 100 million instructions. The score is mispredictions per thousand instructions (MPKI): 6.311 means about six wrong direction guesses in every thousand instructions. "Always taken" guesses that every conditional branch is taken. "gshare" is the 32,768-entry predictor shipped with the simulator. "Hybrid" is `pm_predictor`.

Trace	Always taken	gshare	Hybrid
164.gzip	70.025	12.473	14.543
175.vpr	85.312	13.415	13.693
176.gcc	86.071	11.254	9.126
181.mcf	102.497	15.837	16.920
186.crafty	48.651	5.837	5.129
197.parser	78.421	10.008	8.995
201.compress	76.983	7.831	8.978
202.jess	92.343	1.562	1.148
205.raytrace	109.506	2.756	2.139
209.db	91.451	3.909	3.824
213.javac	109.644	2.267	2.382
222.mpegaudio	110.339	2.188	2.400
227.mtrt	112.022	2.657	2.314
228.jack	77.958	3.033	2.538
252.eon	24.226	1.807	1.730
253.perlbmk	66.833	2.554	1.905
254.gap	75.950	3.926	4.554
255.vortex	70.524	1.222	0.854
256.bzip2	178.257	0.094	0.071
300.twolf	59.221	21.489	22.973
Average	86.311	6.306	6.311

Table 1: Wrong direction guesses per thousand instructions.

Always taken averages 86.311. The hybrid averages 6.311, and gshare averages 6.306. The two accurate predictors differ by 0.005. gcc and perlbmk are better on the hybrid. gzip and twolf are better on gshare.

The accuracy matches because the global table is reserved for branches the local table gets wrong. A stored global guess therefore does not replace a local guess that was already right.

	gshare	Hybrid
Stored bits	65,551	30,735
Physical rows	256	64 and 64
Average MPKI	6.306	6.311

Table 2: Size of the two predictors. Physical rows are after the bits are folded into a near-square array. One row is read.

gshare stores 32,768 two-bit counters and a 15-bit history, 65,551 bits. The hybrid stores 4,096 local counters (8,192 bits), 2,048 global entries with a 6-bit tag, a 2-bit counter, and a valid bit

(18,432 bits), a 2-bit age on each entry (4,096 bits), and the same 15-bit history. That is 30,735 bits. gshare holds 2.13 times as much state.

A wrong guess costs the same in either design, because the miss rates match. Area and idle power follow the stored bits, so both are 2.13 times lower for the hybrid.

Lookup delay is estimated for a Pentium M Banias, the 130 nm part at 1.3 GHz. One cycle is 769 ps. The fanout-of-4 inverter delay used here is 25.2 ps, so a cycle is about 30 FO4.[1] The memories are not one entry per row. The same bits are folded so the array is near a square, and the address bits that no longer pick a row become a column multiplexer. gshare's 64 Kbit array is 256 rows by 256 columns, with a 128-to-1 column mux. The local table is 64 rows by 128 columns, with a 64-to-1 mux. Each global way is its own 64-row memory, and the four ways are read together.

The terms, in FO4 delays, are a parallel XOR $T_{\oplus} = 2$, a decoder $T_{\mathrm{dec}}(n) = 5+(n-6)$ for an n-bit row address, a bitline and sense amplifier $T_{\mathrm{bit}}(R) = 2.4 \times R/64$, a column mux $T_{\mathrm{col}}(k) = \frac{1}{2}\log_2 k$, four 6-bit tag comparisons $T_{\mathrm{cmp}} = 3$, a way mux $T_{\mathrm{way}} = 2$, and the final local-or-global mux $T_{\mathrm{sel}} = 2$. One row is driven. T_{bit} grows with the number of rows because the idle rows still load the bitline.

$$\begin{aligned} L_{\mathrm{gshare}} &= T_{\oplus} + T_{\mathrm{dec}}(8) + T_{\mathrm{bit}}(256) + T_{\mathrm{col}}(128) \\ &= 2 + 7 + 9.6 + 3.5 = 22.1 \text{ FO4}, \qquad (1) \\ L_{\mathrm{local}} &= T_{\mathrm{dec}}(6) + T_{\mathrm{bit}}(64) + T_{\mathrm{col}}(64) \\ &= 5 + 2.4 + 3 = 10.4 \text{ FO4}, \qquad (2) \\ L_{\mathrm{global}} &= T_{\oplus} + T_{\mathrm{dec}}(6) + T_{\mathrm{bit}}(64) + T_{\mathrm{col}}(8) + T_{\mathrm{cmp}} + T_{\mathrm{way}} \\ &= 2 + 5 + 2.4 + 1.5 + 3 + 2 = 15.9 \text{ FO4}, \qquad (3) \\ L_{\mathrm{hybrid}} &= \max(L_{\mathrm{local}}, L_{\mathrm{global}}) + T_{\mathrm{sel}} \\ &= 15.9 + 2 = 17.9 \text{ FO4}. \qquad (4) \end{aligned}$$

At 25.2 ps per FO4 this is 557 ps for gshare and 451 ps for the hybrid. Both fit in the 769 ps cycle. The hybrid lookup is 4.2 FO4, about 106 ps, shorter. The difference is the bitline: 9.6 FO4 at 256 rows against 2.4 FO4 at 64 rows. The tag compare and the two muxes add 7 FO4, which does not use up that whole saving. This is a sum of those assumptions, not a measured silicon delay.

[1] Banias clock and FO4 delay from the Stanford CPU DB. The gate delays below are taken from a 130 nm measurement of a 64-row memory: a 6-bit decoder plus wordline is 5 FO4, and the bitline plus sense amplifier is 2.4 FO4.

5 Conclusion

The worksheet predictor matches the Appendix B example on all 22 rows. The trace predictor, `pm_predictor`, is the Pentium M hybrid at full size: 4,096 local counters, 15 bits of outcome history, and 512 sets of four tagged counters. On the traces it cuts the always-taken miss rate from 86.311 to 6.311 MPKI, level with gshare at 6.306, in 2.13 times less storage. On the Banias estimate the lookup is 17.9 FO4 against 22.1 FO4 for gshare, about 106 ps inside the same cycle.